

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Cloud”

Project Tile – “Agentic AI for Personalized Course Pathways ”

Domain of Project – Education

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Problem Statement - Agentic AI for Personalized Course Pathways

The Challenge :-

Students often struggle to identify the right learning path because of the vast number of online courses and the lack of personalized guidance. Most learning platforms provide generic recommendations that do not consider a student's interests, career goals, current skill level, learning preferences, or progress. As a result, learners face confusion, skill gaps, low motivation, and inefficient learning, making it difficult to achieve their career aspirations.

The Objective :-

Build an AI-powered Agentic Learning Coach that provides personalized learning guidance and career support. The solution should:

-  **AI Skill Assessment** – Evaluate current skills and identify learning gaps.
-  **Smart Course Recommendations** – Suggest relevant learning resources based on interests and goals.

-  **AI Learning Coach** – Provide personalized mentoring and answer learning-related queries.
-  **AI Study Planner** – Generate daily and weekly study schedules.
-  **Project & Certification Suggestions** – Recommend hands-on projects and industry certifications.
-  **Career Guidance** – Suggest career paths, internships, and job opportunities.
-  **Interview & Resume Support** – Assist with interview preparation and resume skill-gap analysis.
-  **Progress Tracking** – Monitor learning progress, achievements, and study streaks.
-  **Smart AI Recommendations** – Adapt suggestions based on performance and learning history.
-  **Learning Analytics Dashboard** – Visualize study hours, skill growth, productivity, and career readiness.

Proposed Solution

LearnMate AI is an **AI-powered Agentic Learning Coach** that helps students make informed learning and career decisions through personalized guidance and intelligent recommendations. By leveraging **IBM watsonx.ai Granite Models**, the platform understands each learner's interests, career aspirations, skill level, and learning preferences to provide a tailored learning experience. It continuously adapts its recommendations based on user progress, identifies skill gaps, and proactively suggests the next best learning steps. The solution aims to make learning more structured, engaging, accessible, and career-focused while improving employability through AI-driven insights.

AI Agents in LearnMate AI :-

-  **Student Profile Agent** – Collects user interests, career goals, learning preferences, and skill level
-  **Skill Assessment Agent** – Evaluates current knowledge and identifies skill gaps.
-  **Course Recommendation Agent** – Recommends personalized learning resources and courses.
-  **Study Planner Agent** – Creates optimized daily and weekly study schedules.

-  **Career Guidance Agent** – Suggests career paths, certifications, internships, and job opportunities.
-  **Interview & Resume Agent** – Provides resume skill-gap analysis and interview preparation.
-  **Progress Analytics Agent** – Tracks learning progress, achievements, and productivity trends.
-  **Adaptive Recommendation Agent** – Continuously updates recommendations based on user performance and learning history.
-  **AI Mentor Agent** – Delivers personalized mentoring, motivation, and answers learning-related queries.

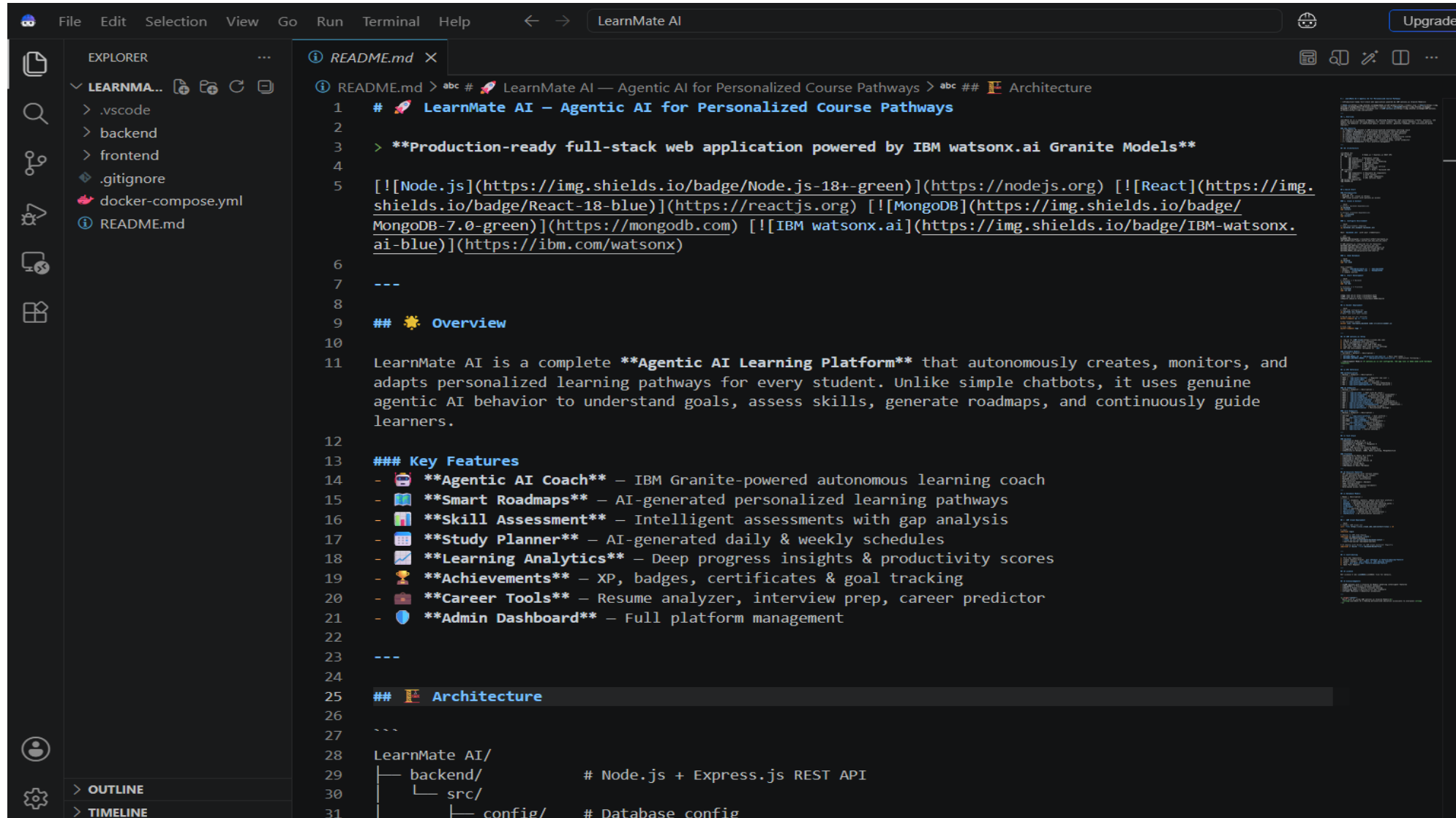
Together, these AI agents work collaboratively to deliver a personalized, adaptive, and intelligent learning experience. Powered by IBM watsonx.ai Granite Models, LearnMate AI helps students bridge skill gaps, stay motivated, make informed career decisions, and achieve their learning goals efficiently.

Technologies Used

- **IBM Granite 3.2 8B Model** – For AI mentoring, skill assessment, personalized learning recommendations, career guidance, interview preparation, and resume skill-gap analysis.
- **IBM watsonx.ai** – Used to deploy and access Granite foundation models for AI-powered learning assistance, roadmap generation, and career recommendations.
- **IBM Cloud** – Provides a secure, scalable, and cloud-native platform for deploying and managing the LearnMate AI application.
- **React (Vite) & Tailwind CSS** – Used to build a modern, responsive, and interactive user interface
- **Node.js & Express.js** – Used to develop secure and scalable REST APIs for backend services
- **MongoDB** – Stores user profiles, learning progress, assessments, goals, chat history, and application data.

- **JWT & bcrypt** – Provide secure user authentication, authorization, and password encryption.
- **Chart.js / Recharts** – Used to visualize learning progress, study analytics, productivity trends, and career readiness.
- **Docker** – Enables containerized deployment for portability, scalability, and easier application management.
- **REST APIs** – Facilitate seamless communication between the frontend, backend, database, and IBM watsonx.ai services.

Code Implementation



```
1 # 🚀 LearnMate AI — Agentic AI for Personalized Course Pathways
2
3 > **Production-ready full-stack web application powered by IBM watsonx.ai Granite Models**
4
5 [[!Node.js](https://img.shields.io/badge/Node.js-18+-green)](https://nodejs.org) [[!React](https://img.shields.io/badge/React-18-blue)](https://reactjs.org) [[!MongoDB](https://img.shields.io/badge/MongoDB-7.0-green)](https://mongodb.com) [[!IBM watsonx.ai](https://img.shields.io/badge/IBM-watsonx.ai-blue)](https://ibm.com/watsonx)
6
7 ---
8
9 ## 🌟 Overview
10
11 LearnMate AI is a complete Agentic AI Learning Platform that autonomously creates, monitors, and adapts personalized learning pathways for every student. Unlike simple chatbots, it uses genuine agentic AI behavior to understand goals, assess skills, generate roadmaps, and continuously guide learners.
12
13 ### Key Features
14 - 🧑‍🏫 Agentic AI Coach — IBM Granite-powered autonomous learning coach
15 - 📖 Smart Roadmaps — AI-generated personalized learning pathways
16 - 📊 Skill Assessment — Intelligent assessments with gap analysis
17 - 📅 Study Planner — AI-generated daily & weekly schedules
18 - 📈 Learning Analytics — Deep progress insights & productivity scores
19 - 🏆 Achievements — XP, badges, certificates & goal tracking
20 - 📄 Career Tools — Resume analyzer, interview prep, career predictor
21 - 🛠️ Admin Dashboard — Full platform management
22
23 ---
24
25 ## 🏗️ Architecture
26
27 ---
28 LearnMate AI/
29 |— backend/      # Node.js + Express.js REST API
30 |   |— src/
31 |   |— config/   # Database config
```

```
File Edit Selection View Go Run Terminal Help LearnMate AI Upgrade your VS Code

EXPLORER
LEARNMATE...
  .vscode
  backend
    logs
    node_modules
    src
  .env
  .env.example
  Dockerfile
  package-lock.json
  package.json
  frontend
    node_modules
    src
  Dockerfile
  index.html
  nginx.conf
  package-lock.json
  package.json
  JS postcss.config.js
  JS tailwind.config.js
  vite.config.js
  .gitignore
  docker-compose.yml
  README.md

tailwind.config.js
frontend > JS tailwind.config.js > ...
1 /** @type {import('tailwindcss').Config} */
2 export default {
3   content: ['./index.html', './src/**/*.js,ts,jsx,tsx'],
4   theme: {
5     extend: {
6       colors: {
7         primary: { 50: '#eff6ff', 100: '#dbeafe', 200: '#bfdbfe', 300: '#93c5fd', 400: '#60a5fa', 500: '#3b82f6', 600: '#2563a2', 700: '#1e408c', 800: '#1c4587', 900: '#1a3d54' },
8         secondary: { 500: '#8b5cf6', 600: '#7c3aed', 700: '#6d28d9' },
9         accent: { 500: '#06b6d4', 600: '#0891b2' },
10        success: { 500: '#22c55e', 600: '#16a34a' },
11        warning: { 500: '#f59e0b', 600: '#d97706' },
12        danger: { 500: '#ef4444', 600: '#dc2626' },
13      },
14      fontFamily: { sans: ['-apple-system', 'BlinkMacSystemFont', 'Segoe UI', 'Roboto', 'sans-serif'] },
15      animation: {
16        'fade-in': 'fadeIn 0.5s ease-in-out',
17        'slide-up': 'slideUp 0.4s ease-out',
18        'pulse-slow': 'pulse 3s infinite',
19      },
20      keyframes: {
21        fadeIn: { '0%': { opacity: '0' }, '100%': { opacity: '1' } },
22        slideUp: { '0%': { transform: 'translateY(20px)', opacity: '0' }, '100%': { transform: 'translateY(0)', opacity: '1' } },
23      },
24    },
25  },
26  plugins: [],
27 }
28
```

```
File Edit Selection View Go Run Terminal Help LearnMate AI Upgrade your VS Code

EXPLORER
LEARNMATE...
  .vscode
  backend
    logs
    node_modules
    src
  .env
  .env.example
  Dockerfile
  package-lock.json
  package.json
  frontend
  .gitignore
  docker-compose.yml
  README.md

.env.example
backend > .env.example
14 # IBM watsonx.ai
15 WATSONX_API_KEY=your_ibm_watsonx_api_key
16 WATSONX_PROJECT_ID=your_ibm_watsonx_project_id
17 WATSONX_URL=https://us-south.ml.cloud.ibm.com
18 WATSONX_MODEL_ID=ibm/granite-13b-chat-v2
19 WATSONX_INSTRUCT_MODEL=ibm/granite-13b-instruct-v2
20 WATSONX_CODE_MODEL=ibm/granite-20b-code-instruct-v2
21
22 # Email (Nodemailer)
23 EMAIL_HOST=smt.gmail.com
24 EMAIL_PORT=587
25 EMAIL_USER=your_email@gmail.com
26 EMAIL_PASS=your_email_password
27 EMAIL_FROM=LearnMate AI <noreply@learnmate.ai>
28
29 # Frontend URL (CORS)
30 FRONTEND_URL=http://localhost:5173
31
32 # Rate Limiting
33 RATE_LIMIT_WINDOW_MS=900000
34 RATE_LIMIT_MAX=100
35
36 # File Upload
37 MAX_FILE_SIZE=5242880
38 UPLOAD_PATH=./uploads
39
40 # Admin
41 ADMIN_EMAIL=admin@learnmate.ai
42 ADMIN_PASSWORD=Admin@123456
43
```

Project Demonstration

 **LearnMate AI**

FeaturesHow It WorksReviews

Login

Get Started →

Your AI-Powered Learning Coach

LearnMate AI autonomously creates, monitors, and adapts your personalized learning pathway using genuine Agentic AI — not just a chatbot.

[Start Learning Free](#) →[Watch Demo](#) >

✔ Free to start

✔ No credit card

✔ Cancel anytime

50K+

Active Learners

95%

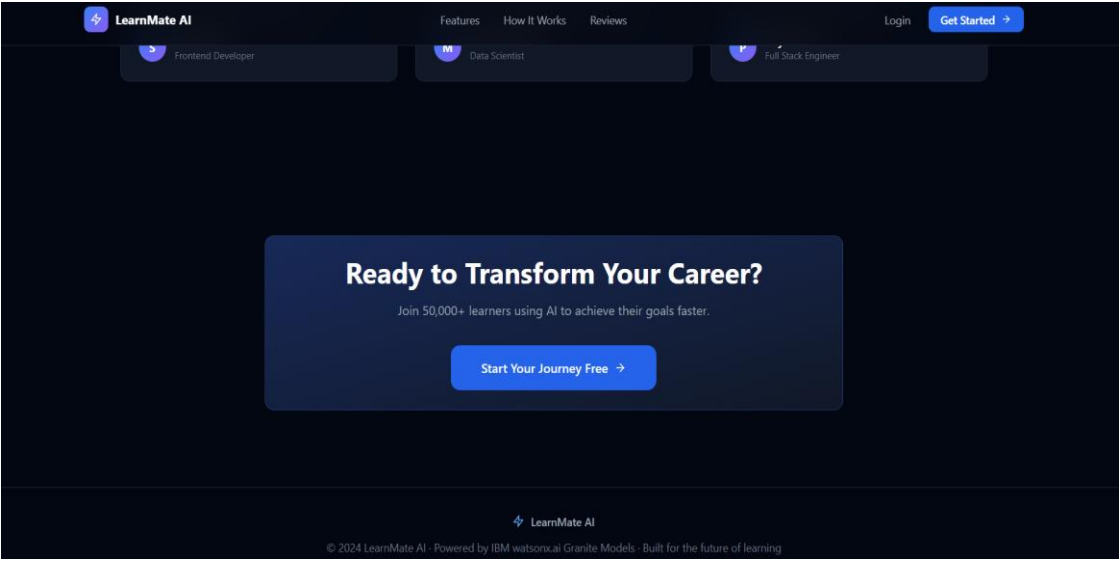
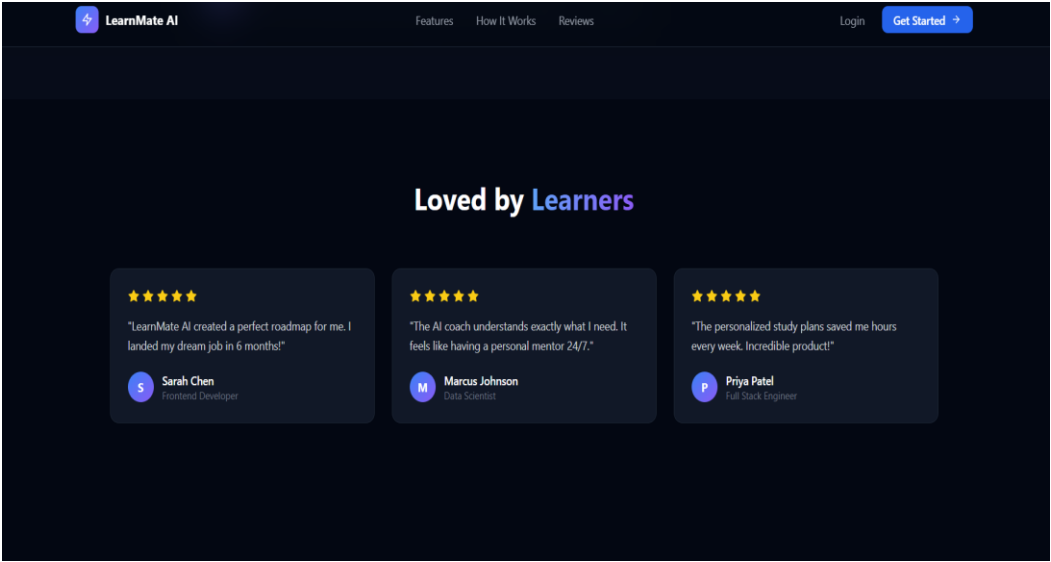
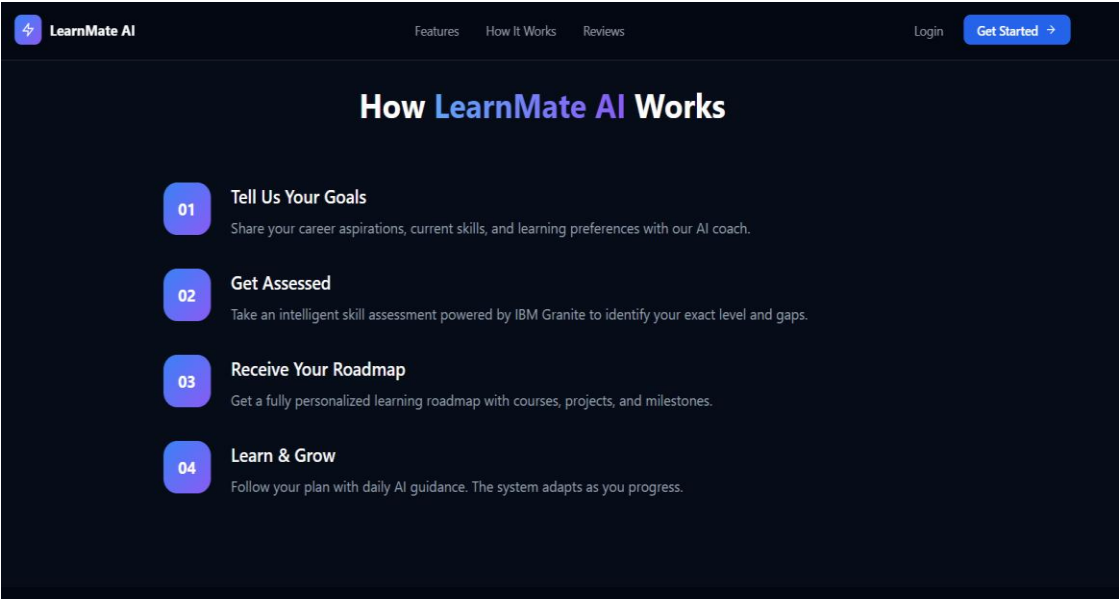
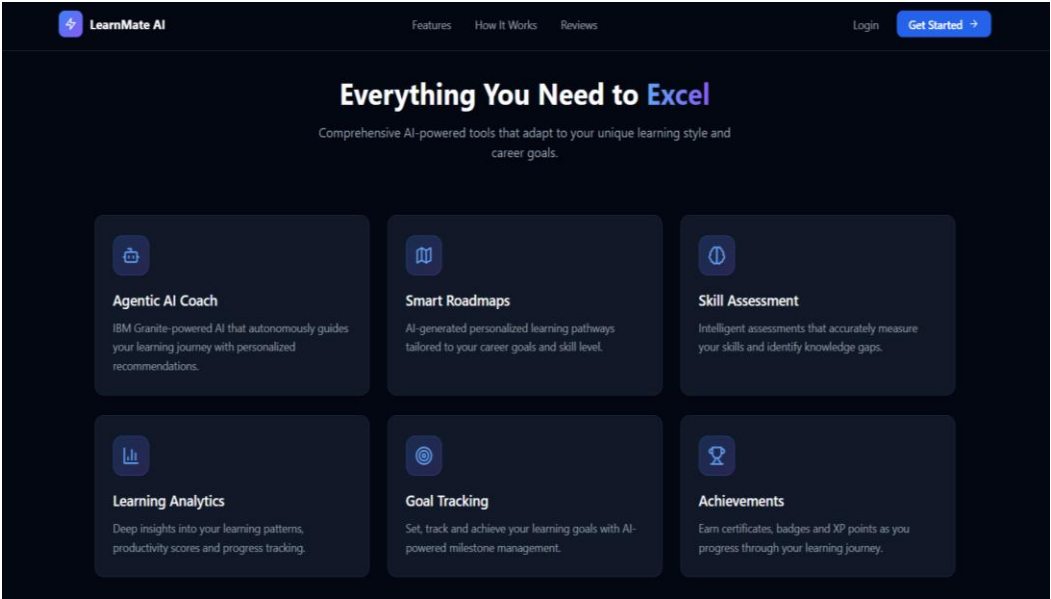
Goal Achievement Rate

200+

Learning Pathways

4.9★

Average Rating



User Login & Registration



Start Your Learning Journey Today

Join thousands of learners using AI to accelerate their career growth.

- ✓ Personalized AI learning roadmap
- ✓ IBM watsonx.ai Granite coaching
- ✓ Smart skill assessments
- ✓ Progress analytics & certificates

Free forever · No credit card required

Create your account

Get your personalized AI learning roadmap in minutes

Full Name

John Doe

Email

you@example.com

Password

Min 8 characters



Career Goal (optional)

e.g. Become a Full Stack Developer

Create Free Account

Already have an account? [Sign in](#)

Let's personalize your experience

Step 1 of 4

Your Career Goal

What do you want to achieve?

Career Goal

e.g. Become a Full Stack Developer

Target Role

Frontend Developer

Backend Developer

Full Stack Developer

Data Scientist

ML Engineer

DevOps Engineer

UI/UX Designer

Mobile Developer

Cybersecurity Analyst

Product Manager

Data Analyst

Cloud Architect

< Back

Next >

Let's personalize your experience

Step 2 of 4

Your Background

Tell us about your experience

Current Role

e.g. Student, Junior Developer

Education

Select education level

Years of Experience

0

Industry

e.g. Technology

< Back

Next >

Learning Style

How do you learn best?

Learning Style

Visual (Videos)

Reading

Hands-on

Mixed

Daily Study Hours: 2h



Preferred Study Time

Morning

Afternoon

Evening

Night

Flexible

Interests

Web Development

Mobile Apps

AI/ML

Data Science

DevOps

Cybersecurity

UI/UX Design

Backend Dev

Cloud Computing

Blockchain

Game Dev

Product Management

< Back

Next >

Powered by IBM watsonx.ai Granite

Ask AI Coach

D

+ New Conversation

CONTEXT MODE

General

Roadmap

Assessment

Career

RECENT CHATS

Generate a study plan for this week

LearnMate AI Coach

Online - IBM Granite

General

Hi Dolly! I'm your LearnMate AI Coach, powered by IBM watsonx.ai Granite. I'm here to **autonomously guide your learning journey**. I can:

Create personalized learning roadmaps

Assess your skills and identify gaps

Guide your career path

Build weekly study plans

Help with interview prep

What would you like to work on today?

09:38 PM

Generate a study plan for this week

0

09:38 PM

Hello Dolly! I'm excited to help you create a study plan for this week. Given your strong foundation in Machine Learning and your career aspiration of becoming a DevOps Engineer, let's focus on areas that will bridge the gap between these two fields.
Day 1 & 2 (Monday & Tuesday): DevOps Fundamentals: Let's start by understanding the basic concepts of DevOps. This will include understanding its principles, benefits, and how it differs from traditional IT practices. You can use online platforms like Pluralsight or Udemy to find courses on DevOps fundamentals.
Day 3 (Wednesday): Infrastructure as Code (IaC): This concept is crucial in DevOps and goes hand in hand with your machine learning background. Tools like Terraform and CloudFormation will be

Ask your AI coach anything... (Enter to send)

Powered by IBM Granite

AI responses are generated by IBM watsonx.ai Granite. Always verify important information.

LearnMate AI

Agentic Learning

Powered by IBM watsonx.ai Granite

D

Dolly

0 XP - Lvl 1

Dashboard

AI Coach

Assessment

Roadmap

Study Planner

Analytics

Achievements

Profile

Settings

Logout

Start New Assessment

Select Domain

Web Development

Data Science

Machine Learning

DevOps

Mobile Development

UI/UX Design

Cybersecurity

Cloud Computing

Blockchain

Game Development

Your Level

Beginner

New to this field

Intermediate

1-3 years exp

Advanced

3+ years exp

AI-Generated Questions

10 personalized questions · ~20 minutes · Instant analysis

Start Assessment

Powered by IBM watsonx.ai Granite

Ask AI Coach

D

Study Planner

AI-generated weekly schedule tailored to your goals

Log Study

Regenerate

Week

Week 1

Total Hours

8h

Daily Goal

2h/day

Completed

0 tasks

Stay consistent and patient with yourself. Learning is a journey that allows you to grow daily.

Weekly Overview

Mon

2

Tue

2

Wed

2

Thu

2

Fri

2

Weekly Milestones

Complete initial assessment

Gain familiarity with all subjects

Begin project planning

Powered by IBM watsonx.ai Granite

Ask AI Coach

D

Learning Analytics

Deep insights into your learning journey

7d

30d

90d

TOTAL HOURS

0h

0.0h this period

PRODUCTIVITY SCORE

50%

stable

CAREER READINESS

0%

Industry benchmark

LEARNING STREAK

0 days

Best: 0 days

Daily Study Hours

Last 30 days

Skill Radar

Assessment-based

Machine Learning

JavaScript

Python

DevOps

AWS

React

Nodejs

Java

Docker

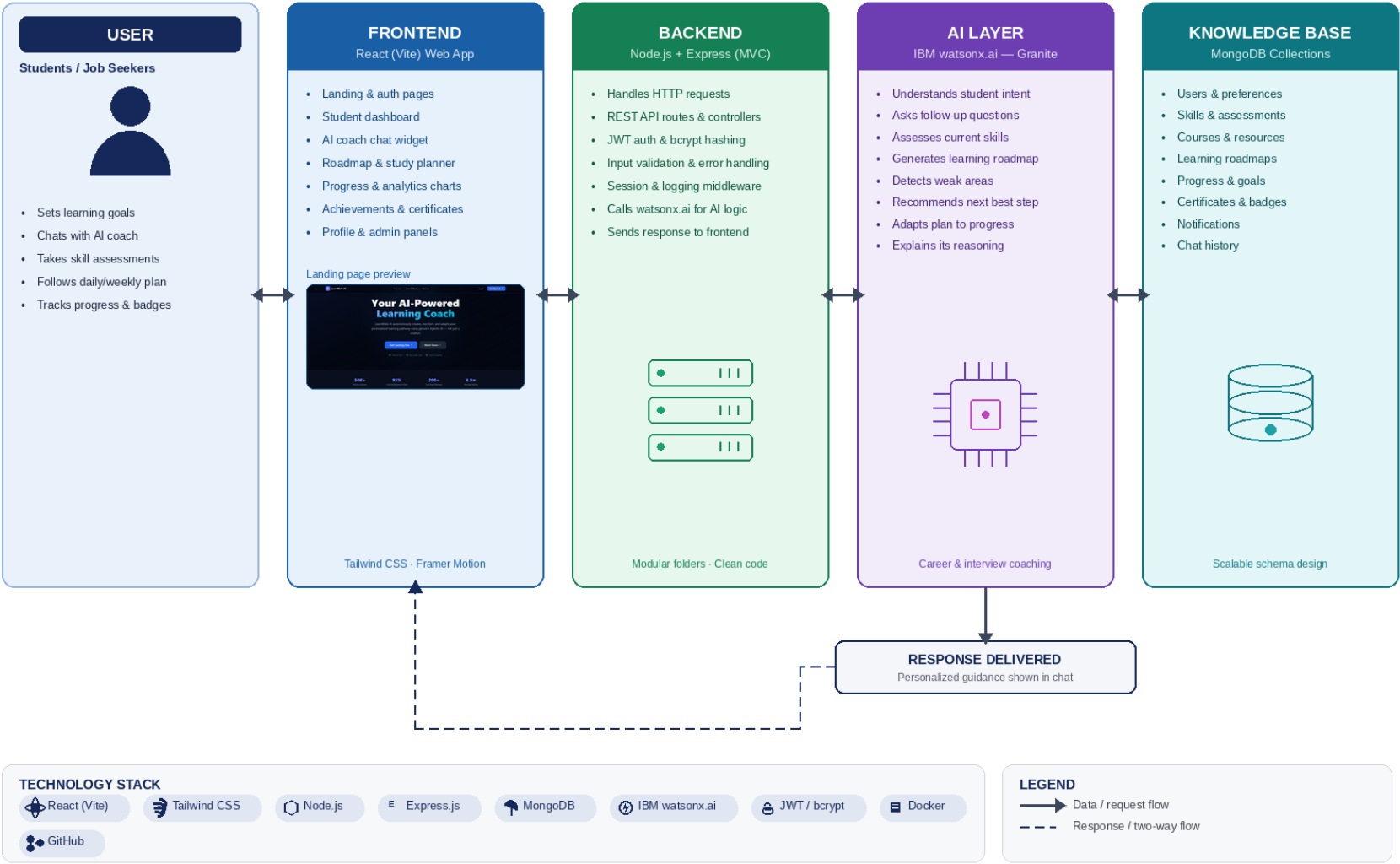
Weekly Study Hours

Goal Progress

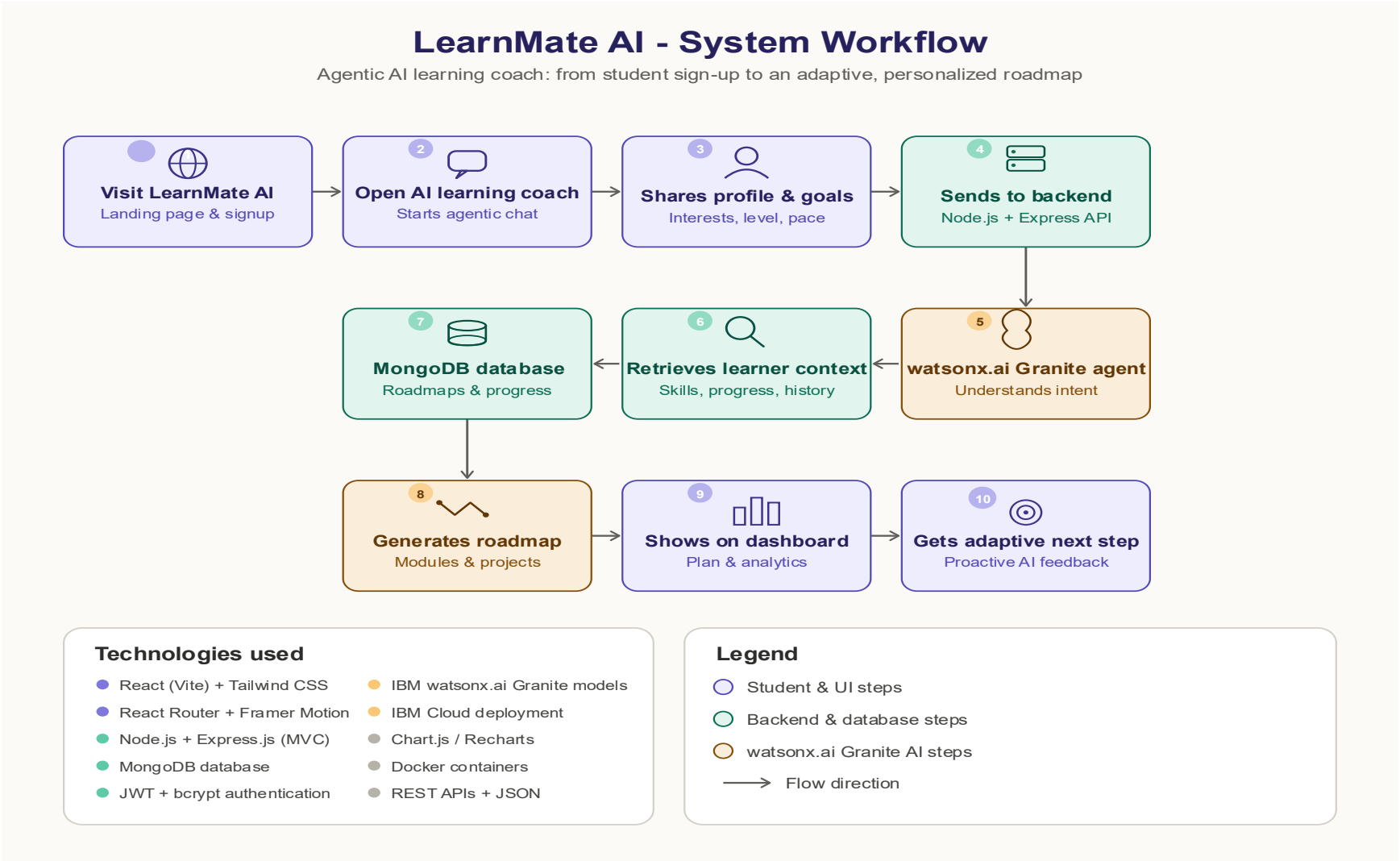
Architecture Blueprint

LEARNMATE AI — SYSTEM ARCHITECTURE

Agentic AI Learning Coach powered by IBM watsonx.ai Granite Models



System Workflow



Role of Agentic AI in LearnMate AI

Agentic AI enables **LearnMate AI** to go beyond a traditional chatbot by acting as an intelligent learning coach that proactively assists students throughout their learning journey. Instead of only responding to queries, it understands user goals, analyzes progress, makes decisions, and continuously adapts recommendations to improve learning outcomes.

- **Student Profile Agent** – Understands learner interests, goals, and preferences.
- **Skill Assessment Agent** – Evaluates current skills and identifies learning gaps.
- **Learning Recommendation Agent** – Suggests the most relevant courses, resources, and projects
- **Career Guidance Agent** – Recommends certifications, internships, and career opportunities.

These AI agents work together to deliver an adaptive, personalized, and autonomous learning experience, helping students build the right skills and achieve their career goals.



GitHub Repository :-

- <https://github.com/VatsalyaKV/LearnMate-AI.git>

Novelty and Uniqueness

LearnMate AI uses **Agentic AI** to provide personalized, adaptive, and proactive learning guidance. Powered by **IBM watsonx.ai Granite Models**, it analyzes user goals and skill gaps to deliver intelligent recommendations, making learning more effective and career-focused.

- **Agentic AI Approach** – Goes beyond a traditional chatbot by proactively understanding user needs and making intelligent learning decisions.
- **Personalized Learning Experience** – Delivers recommendations based on interests, skill level, goals, and learning progress
- **Continuous Adaptation** – Updates learning suggestions dynamically as users complete courses and improve their skills.
- **IBM watsonx.ai Integration** – Leverages IBM Granite Models for intelligent mentoring, career guidance, and personalized recommendations.
- **Career-Focused Learning** – Combines skill assessment, study planning, project recommendations, and career guidance in a single AI-powered platform
- **Scalable & Cloud-Ready** – Built with a modular architecture for secure, scalable deployment on IBM Cloud.

This makes LearnMate AI a smart, scalable, and innovative solution for personalized education and career development

Future Scope

➤ **Mobile Application & Multi-Platform Support :-**

The platform can be expanded to Android and iOS applications, allowing students to access personalized learning guidance, study plans, and AI mentoring anytime and anywhere for a seamless learning experience.

➤ **LMS & University Integration :-**

LearnMate AI can be integrated with Learning Management Systems (LMS), universities, and online learning platforms to automatically track course progress, synchronize learning activities, and provide institution-wide personalized learning support.

➤ **AI Mentor Matching & Collaborative Learning :-**

Future versions can connect students with mentors and enable collaborative learning communities where learners can participate in group discussions, peer learning, and project collaboration

Certificates Earned:

In recognition of the commitment to achieve
professional excellence



Vatsalya K V

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



Issued on: Jun 24, 2026
Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/5af714b2-57db-4b03-a707-8639d1a61347>





This certificate is presented to

Vatsalya K V

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 29 Jun 2026 (GMT)

Learning hours: 30 mins



This certificate is presented to

Vatsalya K V

for the completion of

**Edunet - University Engagements - AICTE
Internship-2026-27**

(PLAN-6B9E5C210607)

According to the Your Learning Builder - Plans system of record

Completion date: 29 Jun 2026 (GMT)

Thank You!

Thank you for your time and interest.